

UNIVERSITY OF SOUTHAMPTON
Faculty of Engineering and Physical Sciences
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A progress report submitted for continuation towards a MEng

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**A Reproducible Evaluation of Linux
Rootkit Detectors via the Implementation
of an Automated Testbed &
Proof-of-Concept Rootkit**

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ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
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This work is all about ...

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Nomenclature

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Acknowledgements

Thanks to no one.

To ...

Chapter 1

Background

1.1 Linux & The Rootkit

Rootkits are typically described as a specialised piece of malware built for persistence & stealth which is installed *after* an attacker has gained root-level access to a system [1]. They may also give an attacker additional functionality such as network backdoors (cite magic bpf), the ability to hide files or other capabilities that give the attacker a greater ability to hide their presence whilst maintaining persistence.

Linux & subsequently Linux rootkits have been chosen for this project due to its relevance and prevalence in cloud & server environments. Additionally, a wide range of rootkit techniques exist for Linux which will be covered in the following section. Consequently, many tools have been developed which aim to detect the presence of a rootkit on a system. To achieve this, these tools may take a variety of approaches which will be covered in section 1.3.

The reliability and accuracy of these tools against current rootkits is paramount to ensure that test test...

1.2 Evolution of Rootkit Techniques

Linux rootkits have had an evolution

1.3 Rootkit Detectors

Chapter 2

Literature Review

Academic interest in the evaluation of rootkit detection rises once newer technologies are introduced & techniques are discovered - such as with eBPF & io_uring. Consequently in the last 10 years there is a sizeable amount of documentation on the various methodologies utilised by those wishing to evaluate a rootkit detector.

This literature review will attempt to cover & summarise the existing methodologies in the literature for evaluating rootkit detector effectiveness. Additionally, It will cover the results found in these studies as useful datapoints for comparison to the results found in this project.

Finally, it will also discuss any relevant literature regarding the development of rootkits, aiming to give a deeper technical explanation of their ability to evade detection.

2.1 Existing Methodologies For Evaluating Rootkits

2.2 Results of Current Evaluations

2.3 Modern Rootkit Design & Detector Engineering

Bibliography

- [1] National Institute of Standards and Technology, “Rootkit,” Computer Security Resource Center Glossary, n.d., accessed: 07-10-2026. [Online]. Available: <https://csrc.nist.gov/glossary/term/rootkit>